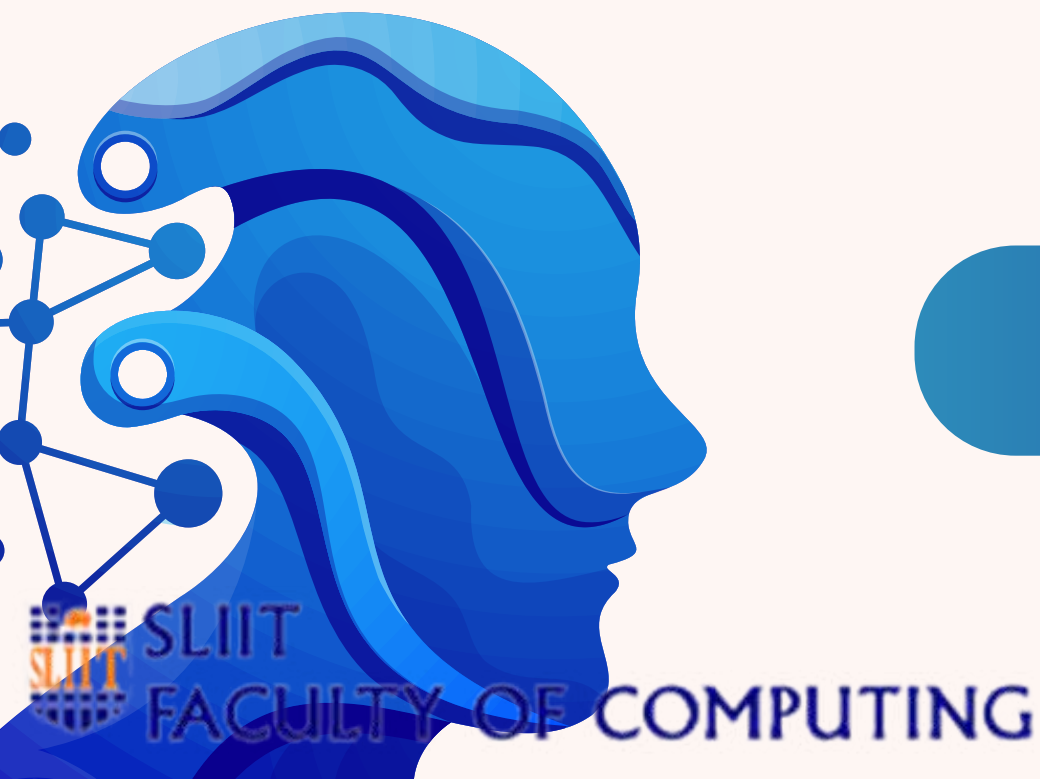


AI-Based Student Wellbeing Monitoring System for Sri Lankan Schools

A Smart System to Support Student Wellbeing and Success

R26-IT-165



Our Team



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Introduction

- Student wellbeing and academic performance are closely connected in school environments.
- Many schools lack effective systems to continuously monitor student stress and learning progress.
- As a result, at-risk students are often identified too late for timely support.
- This research proposes an AI-based platform to monitor student wellbeing, analyze learning data, and support early intervention.

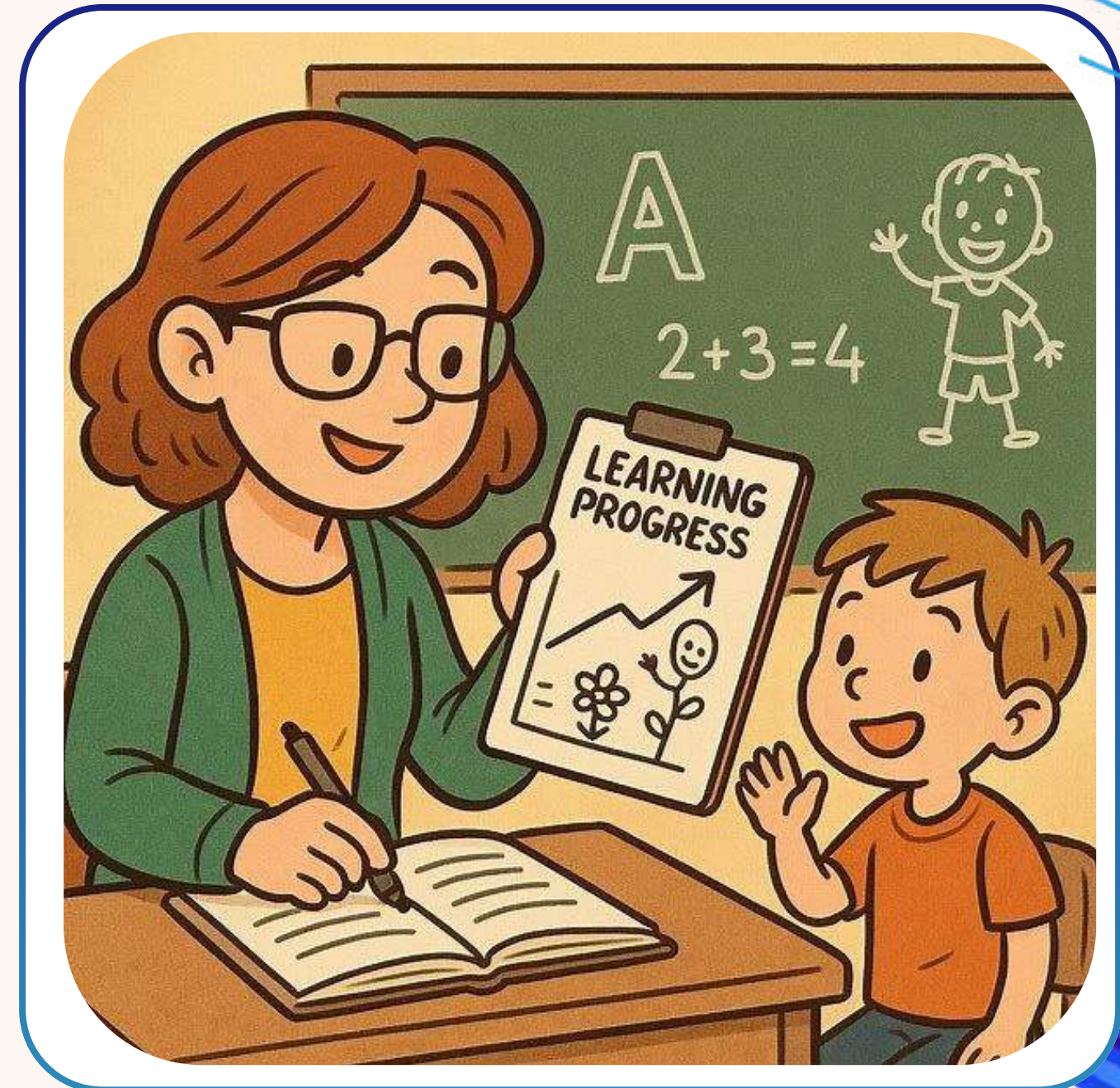
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How can schools effectively detect at-risk students early without an intelligent system to continuously monitor student wellbeing and academic performance?

MAIN OBJECTIVE

To develop an AI-based system that monitors student wellbeing and learning performance in order to detect at-risk students early and support timely intervention in schools.

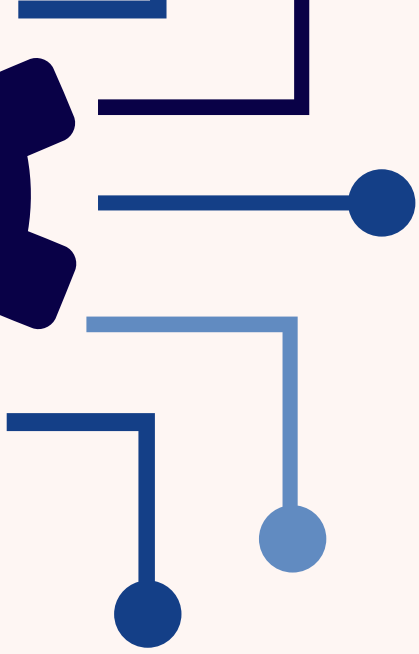


SUB OBJECTIVES

- Wellbeing Indicator Modeling, Data Framework & Learning Content Processing
- AI-Based Risk Analysis Engine
- Monitoring, Visualization & AI Learning Support Dashboard
- coping mechanism recommendations and social peer matching



OVERALL SYSTEM DIAGRAM



Wellbeing Indicator Modeling, Data Framework & Learning Content Processing

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Background

- Schools collect student data such as attendance, exam results, surveys, and teacher observations. However, these data sources are often not structured for effective analysis of student wellbeing and learning progress.
- This component develops measurable wellbeing indicators to better understand students' academic and emotional conditions.
- It also processes teacher guidebook PDFs to extract learning content that supports automated question generation and practice assessments. The structured dataset generated in this component provides the foundation for AI-based analysis and the overall student monitoring system.

Research question

- ❖ How can a structured student wellbeing data framework that integrates wellbeing indicators, student self-assessment surveys, and school academic records improve the accuracy of monitoring student wellbeing and learning performance in educational environments?
- ❖ How can automated processing of teacher guidebook PDFs and learning materials be used to generate structured learning datasets that support intelligent question generation and student answer evaluation systems?



Main Objective



❖ To design and develop a structured data framework that models student wellbeing indicators and processes school data and learning materials to support AI-based analysis, automated question generation, and student learning evaluation.

Sub Objective

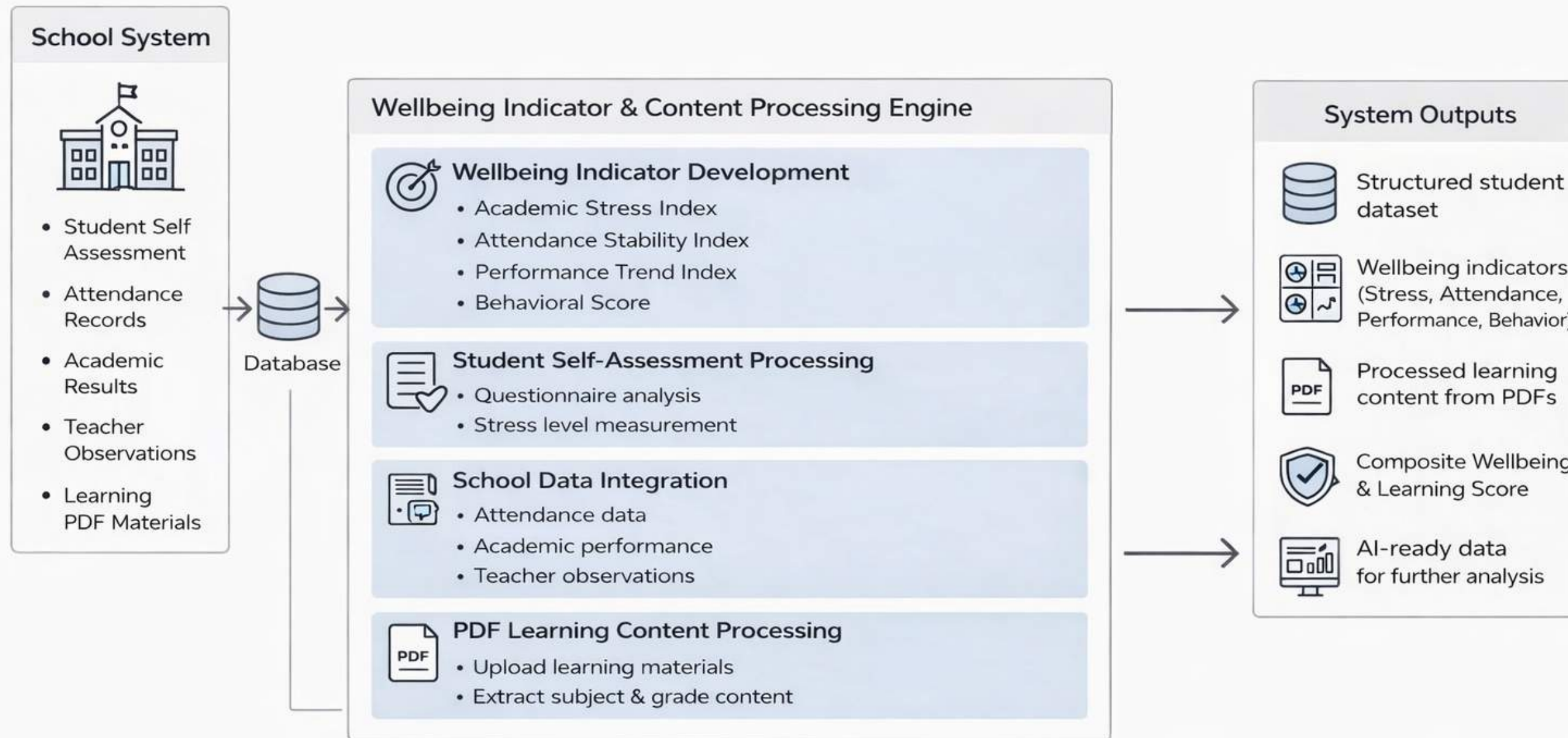
To develop measurable student wellbeing indicators such as stress, attendance, and performance trends.

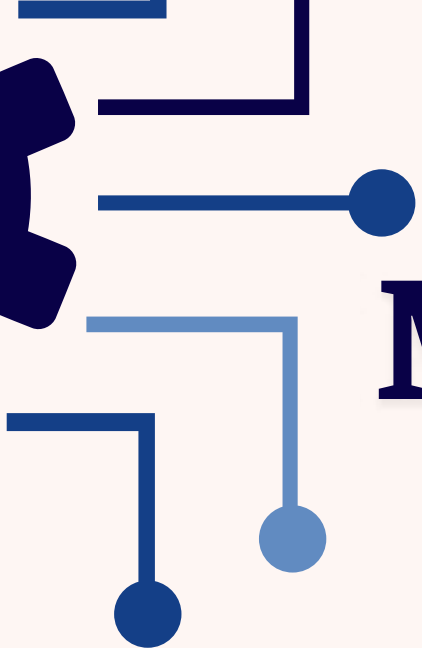
To integrate survey data and school records into a structured student wellbeing dataset.

To process teacher guidebook PDFs to extract learning content for automated question generation and answer evaluation.



Component Diagram





Methodology

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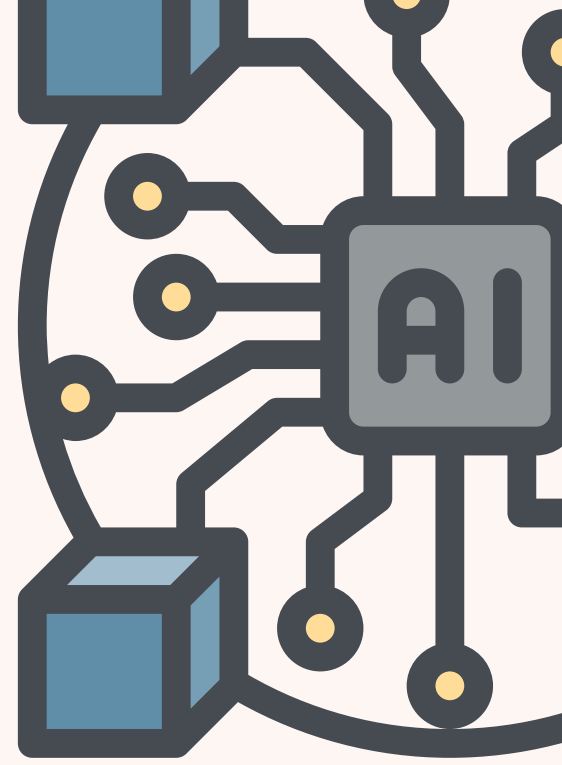
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CONCLUSION

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AI - Based Risk Analysis Engine

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Background

AI-Based Risk Analysis Engine analyzes student data such as attendance, academic performance, behavior, and stress using machine learning to identify students who may be at risk.



Research Questions

1. How can machine learning techniques be used to analyze student wellbeing data to identify at-risk students?
2. Can AI models effectively detect patterns between academic performance, attendance, behavior, and stress indicators?
3. How accurately can an AI model classify students into different wellbeing risk levels?



Main Objectives

To develop an AI-based risk analysis engine that analyzes multidimensional student wellbeing data and predicts students who may be at risk, enabling early identification and timely intervention.

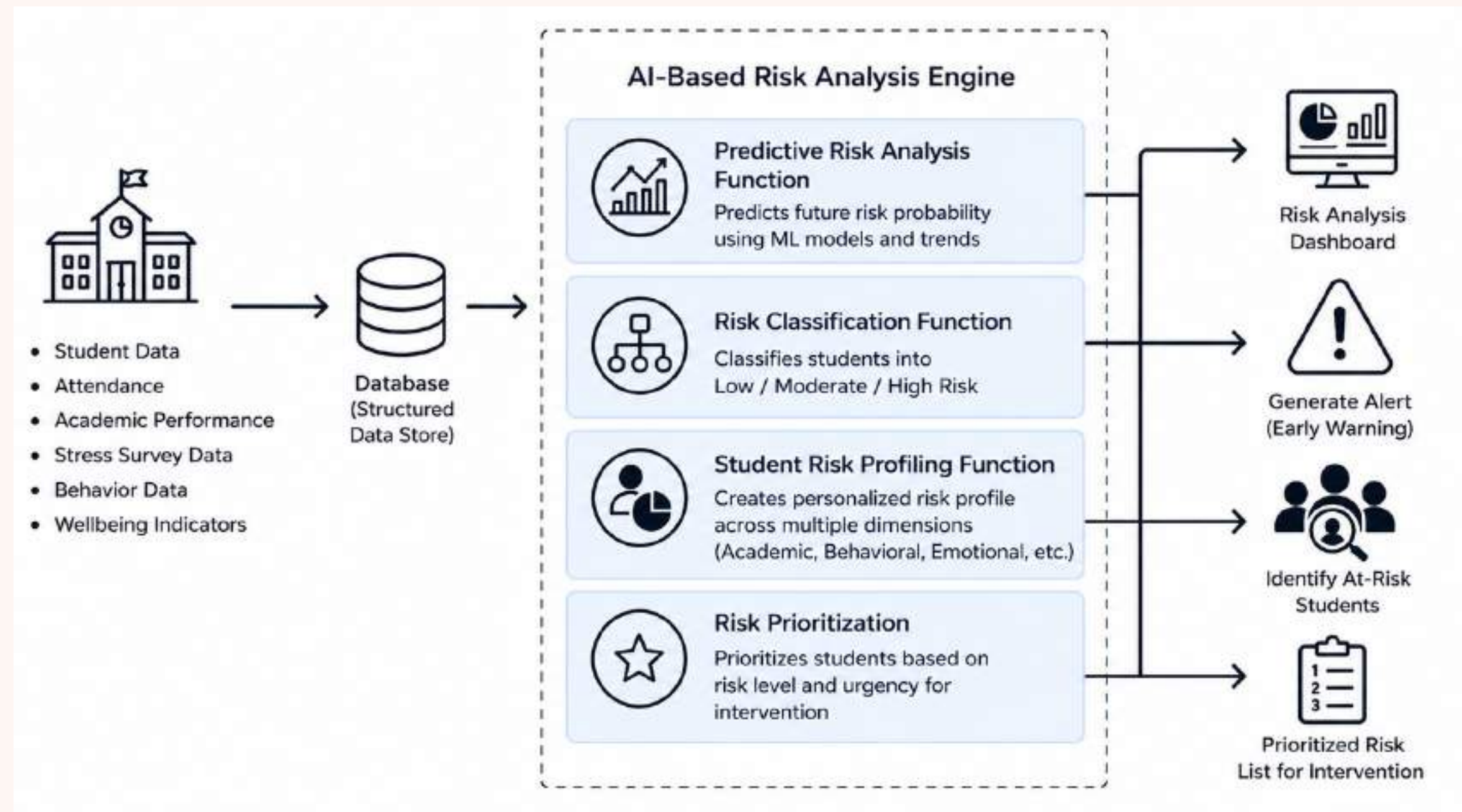
Sub-Objectives

Detect patterns and correlations in student data.

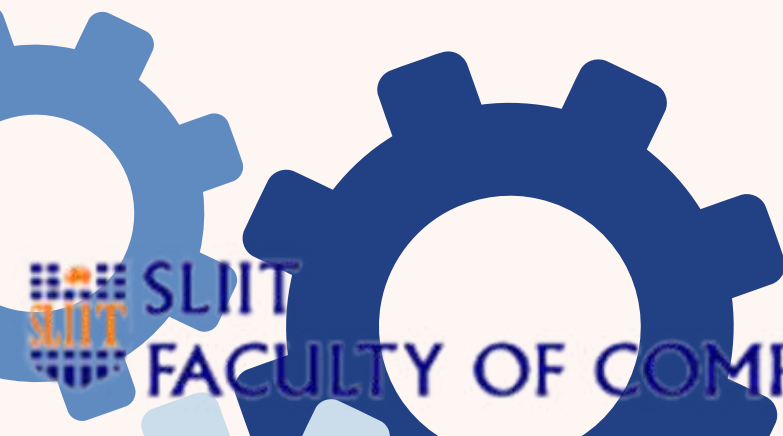
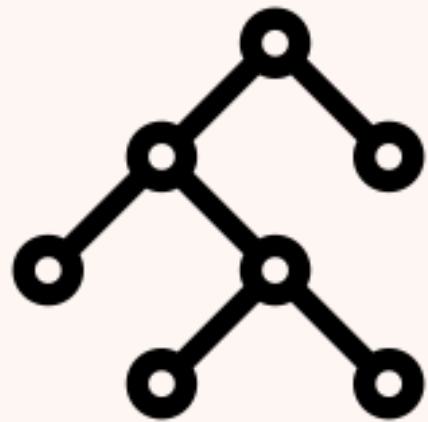
To identify patterns and correlations between different wellbeing factors.

Provide outputs through alerts and a risk analysis dashboard.

Component Diagram



Technologies and Tools



Evidences For the completion



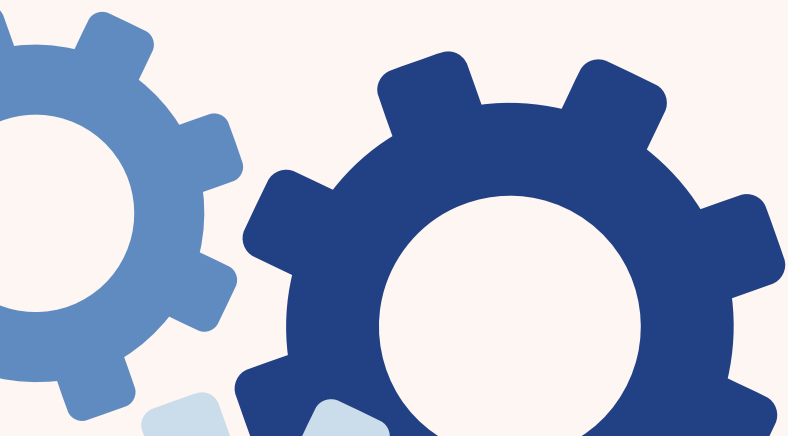
Evidences For the completion



Monitoring, Visualization & AI Learning Support Dashboard

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Background

- Academic

Personalized Coping Recommendations & Social Isolation Detection System

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BACKGROUND

Academic pressure and social transitions have led to a significant surge in student distress. Current support systems are often reactive rather than proactive.

- 14% of students aged 10-16 face mental disorders.
- 40% experience persistent sadness or hopelessness.
- Traditional methods fail to capture micro-social isolation.
- Personalized coping mechanisms are rarely integrated into the learning environment.



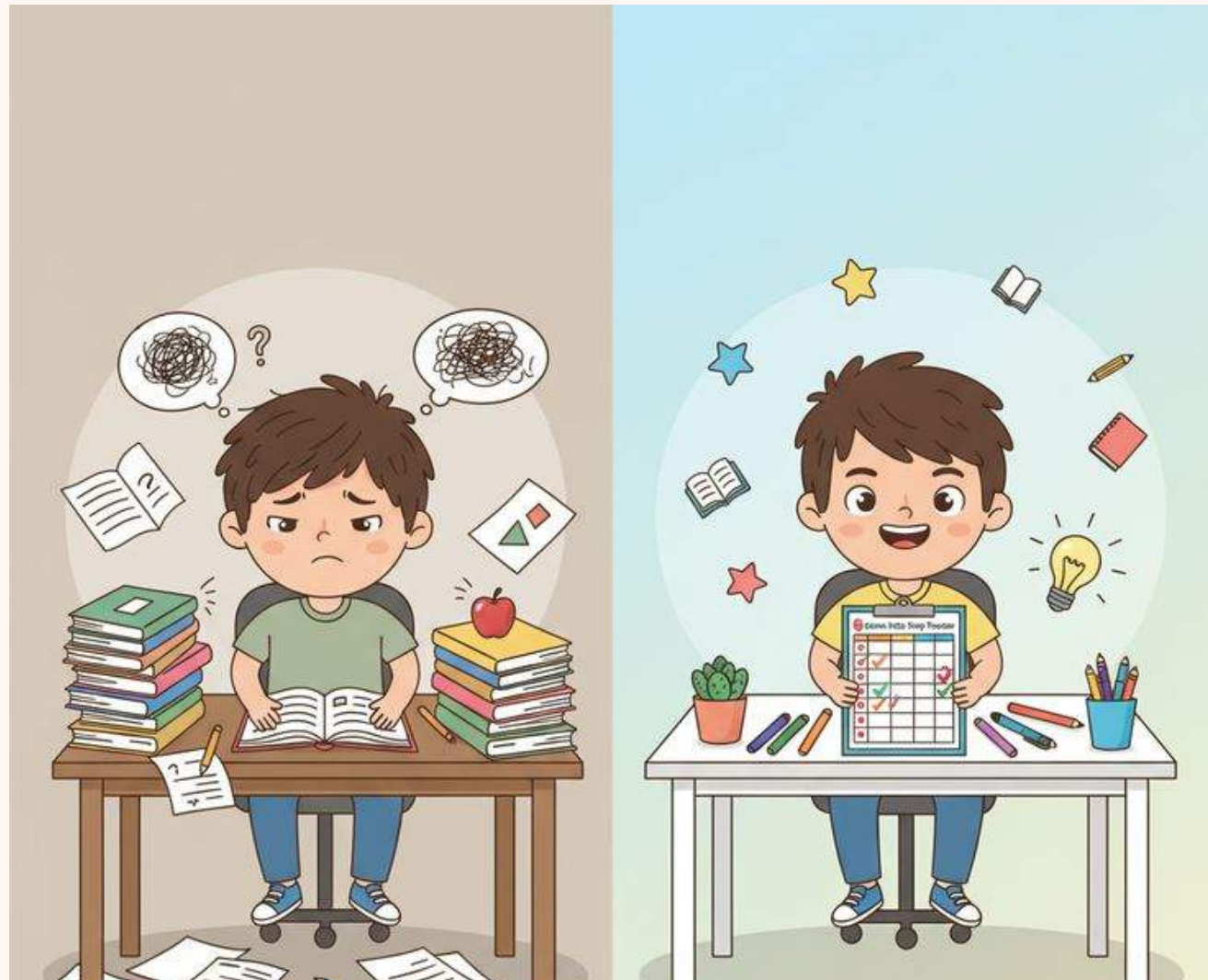
RESEARCH QUESTION

How can a real-time system utilizing sentiment expressions and behavioral clustering personalize stress management and detect social isolation for early educator intervention?

This research explores the intersection of Recommender Systems and Cluster Analysis to bridge the gap between student feelings and teacher action.



MAIN OBJECTIVE



To develop an integrated platform that personalizes coping strategies through machine learning recommendations and classifies student social health using unsupervised clustering for targeted teacher intervention.



SUB OBJECTIVE

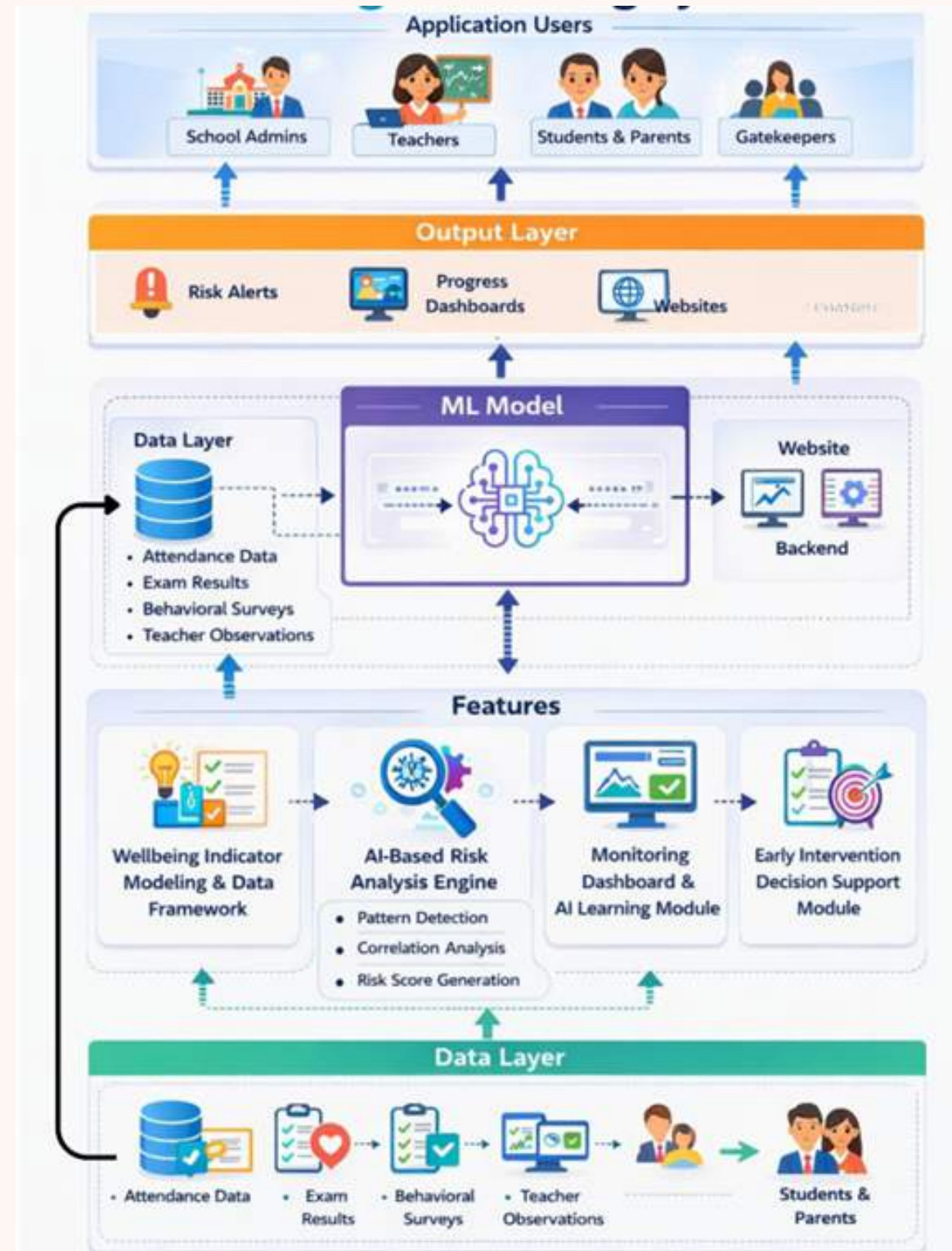
Construct a recommendation engine to identify the top three effective coping mechanisms tailored to individual stress profiles.

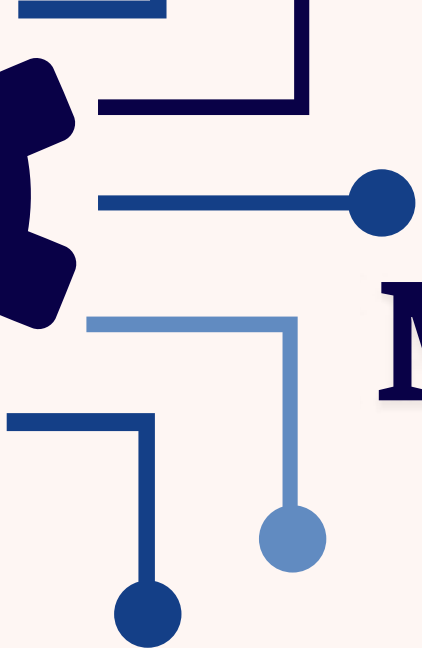
Develop an isolation detection model using facial/textual expressions to group students into Isolation, Moderate, or High Interaction clusters.

Create a dashboard for teachers to implement early intervention and group activity matching based on detected social risks.



COMPONENT DIAGRAM



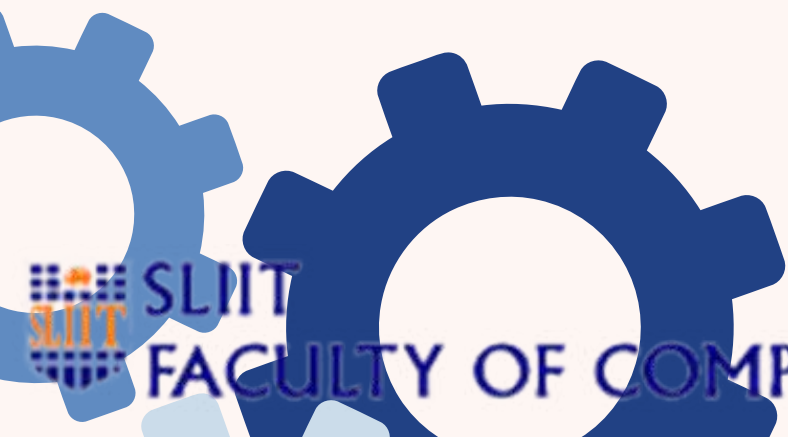
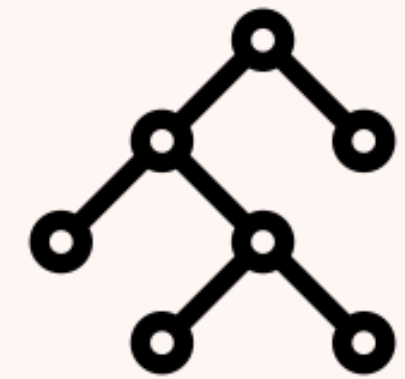


Methodology

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TECHNOLOGIES AND TOOLS



Evidences For the completion

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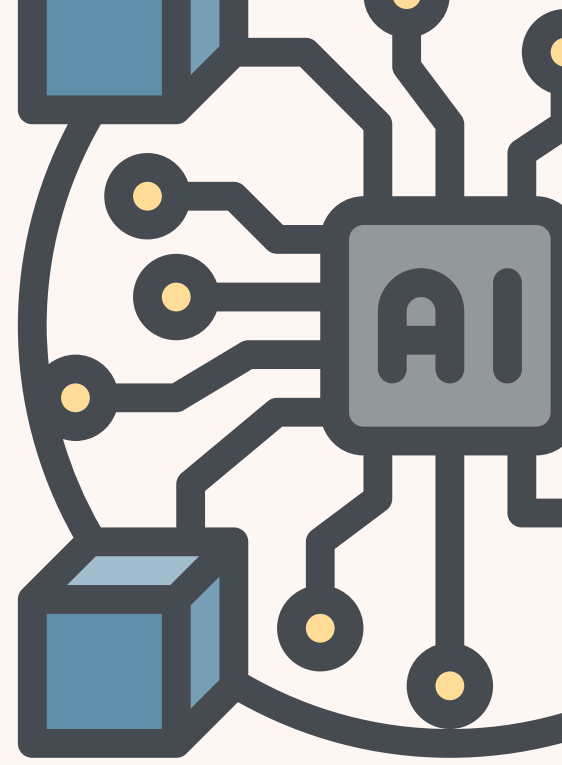
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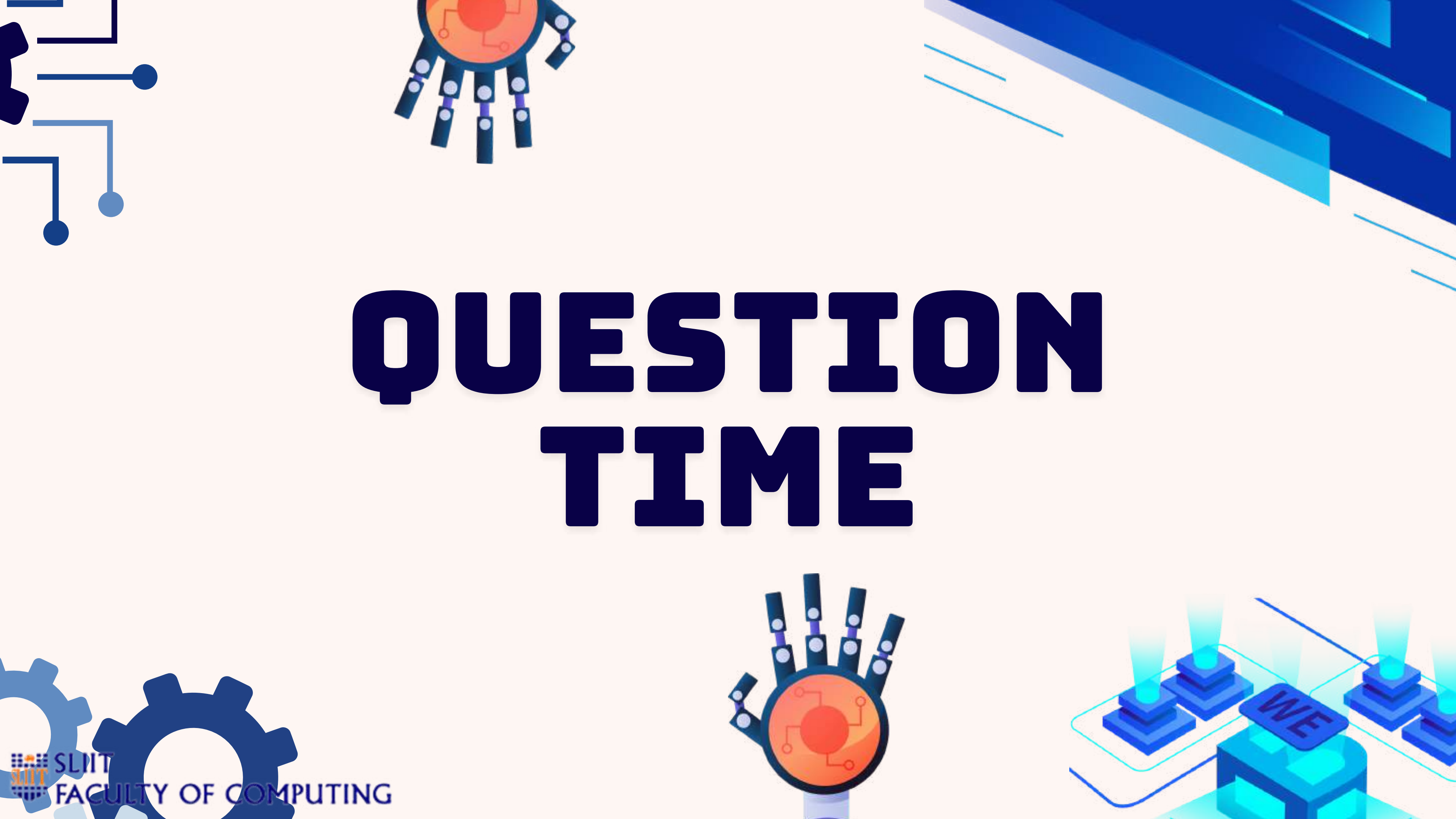


CONCLUSION

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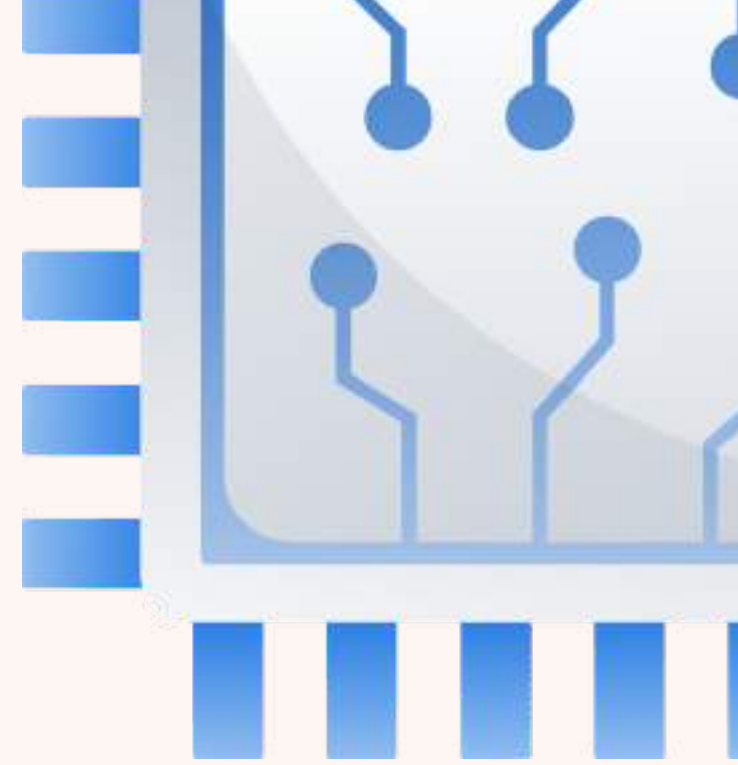
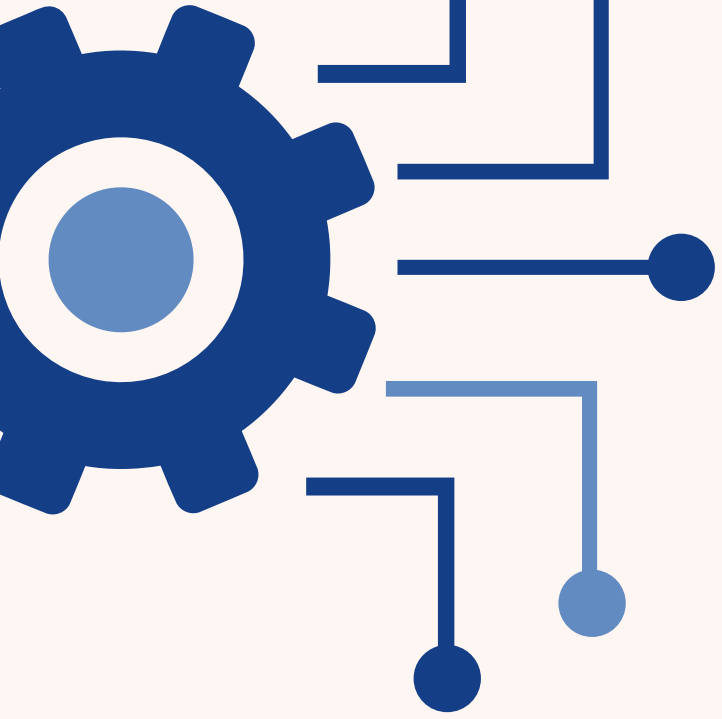
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QUESTION TIME





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